



A global basis of urban tree inventories: What comes first the inventory or the program

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ABSTRACT

While efforts to standardize urban forest inventory data have occurred, the information collected from community to community remains quite variable. In this study, we conducted a global review of peer-reviewed and published research that collected tree inventory data to assess what information is being collected. In addition to this review, we also present results from national surveys conducted in the United States and Sweden that specifically asked urban forest managers about how their inventories informed their urban forest management efforts. In the literature review, tree species and stem diameter were the most collected structural parameters and planning and estimation of ecological services were the most common research objectives. The surveys conducted in Sweden and the USA found communities commonly collected records about tree species, diameter, and condition and that inventories were more common as community size increased. Street and park tree inventories were the most common inventories. Inventories of public trees on greenbelts and woodlots were less common and private tree inventories were relatively uncommon. Additionally, professional staff and consultants were more likely to collect inventory data than interns and volunteers in both countries. Despite these similarities, significant differences exist between the USA and Sweden, with several parameters (i.e., adoption a management plan, presence of an inventory, collection of inventory data digitally, collection of tree planting/removal records) being more common in the USA — possibly reflecting a longer history of urban tree management in the latter country. In modelling the survey results from Sweden and the USA, we found that having a strategic plan, a governance mechanism (tree board), financial resources (per capita budget), and the adoption of greater systematic care of trees were significant predictors the presence of an inventory. Study results also suggest that having a tree inventory reflects the advancement of an urban forestry program along a continuum of initial program adoption to more fully developed and sustained urban forestry program.

1. Introduction

Urban trees have ecological, economic, and social values which can be quantified using data collected during an urban tree inventory (Klobucar et al., 2020; Mcpherson et al., 2016a). Tree inventory systems can depict both the present and past condition of an urban forest and provide basic information for the implementation and refinement of urban forest management methods (Kielbaso, 2008; Long et al., 2008; Miller et al., 2015; Morgenroth et al., 2016). As a decision-making tool, a

tree inventory contains parameter information such as tree species, health status, size, risk level, and location which provide benchmark information for urban forest planning and management (Morgenroth and Östberg, 2017). The parameters selected for a tree inventory are important and should be sufficient to make informed and planned decision making (Miller et al., 2015).

Tree diameter at breast height (DBH; 1.3–1.4 m, depending on regional professional norms) is an important and easy to measure inventory attribute that is commonly collected (Morgenroth et al., 2020).

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While a useful measure in its own regard, tree diameter can also be used to predict other indicators such as tree growth, canopy spread, trunk flare, and tree height (McPherson et al., 2016b; Hilbert et al., 2020). Tree species, condition, insect/disease conditions, and maintenance requirements are also regularly collected (Hauer and Peterson, 2017). In addition to parameters discussed above, long-term monitoring of tree populations ideally includes the date of measurement, a data collector identifier and characterization of experience level, site variables (e.g., land use, spatial location such as global position system (GPS) coordinates), and mortality status (Evarter-Bundere et al., 2014; Roman et al., 2013). Depending on the particular objectives of an urban tree inventory, a sample to estimate the population or a complete inventory (i.e., a census) may be selected (Long et al., 2008).

While in person, ground-based inventories continue to be the norm, imaged-based approaches have been trialed in urban forest settings (Berland and Lange, 2017; Roberts et al., 2019). Differences between field- and image-based methods for collecting tree inventory data exist with both having advantages and limitations (Morgenroth and Östberg, 2017). For example, a field-based assessment can be time consuming to collect data, but should be more accurate since individual trees were visited (Miller et al., 2015). Image-based assessment and data collection require experts, researchers, and/or other trained specialists in the domain of geography but can be effective in assessing large-scale areas when done remotely (Crall et al., 2011). Compared with traditional field measurement, remote sensed imagery combined with a means to detect above-ground structure such as Light Detection and Ranging (LiDAR) can be less costly to collect required information for planning and management depending on the project area, scope, and accuracy requirements (Bastos and Hasegawa, 2013; Chen and Zhu, 2013; Gikunda and Griffith, 2019; Kane and Ryan, 1998; Saarinen et al., 2014). Tree canopy assessment has been used to test the effectiveness of urban forestry policy on land development and retention of tree canopy, as well as the impacts of development and the equity of greenspace areas (Hilbert et al., 2019). Remotely collected inventories can be extremely convenient in terms of obtaining spatial positioning, species diversity, and tree stand conditions over large land areas (Abd-Elrahman et al., 2010; Gikunda and Griffith, 2019; Klobucar et al., 2020; Saarinen et al., 2014; Yépez-Rincón and Lozano-García, 2014). Recently, virtual street tree inventories have used street-level panoramic images collected, processed, and stored online through systems such as Google Street View™ images to assist with data collection (Berland et al., 2019; Berland and Roman, 2020). However, ground-based image reconnaissance was important for verification of a tree inventory and Google currently restricts the use of Street View imagery for tree inventories and research activities.

Tree inventories are typically collected through arborists, urban foresters, and other qualified professionals (Hauer and Peterson, 2017). Engagement and community (citizen) science data which allow the general public to participate in tree data collection have been used with varying success (Hauer et al., 2018; Roman et al., 2017). The accuracy of a tree inventory determines what management decisions may be derived from it (Fischer et al., 2007), and compared with professionals, the accuracy of data measured by public volunteers can be lower which may or may not impact its intended use (Bloniarz and Ryan, 1996; Cozad, 2005).

Tree inventories are useful to create work orders for urban forestry management operations. Tree risk management is an important example of how a tree inventory can be used to manage urban trees (Keller and Konijnendijk, 2012; Koeser et al., 2016; Matthews et al., 2015; Östberg et al., 2018; Terho and Hallaksela, 2005). The tree inventory is used to provide basic data for risk management, that, when applied in operations such as pruning, installation of support systems, or removal, should reduce population-wide tree risk (Abd-Elrahman et al., 2010; Östberg et al., 2018; Pearlmutter et al., 2017). The adoption of systematic risk management activities was related to city size with larger communities more likely to undertake risk management (Koeser et al., 2016). Koeser

et al. (2016) also found that tree risk management programs may develop as a result of a past claim for damage or injury.

Tree inventories can be used to evaluate tree growth and longevity of urban forests, tree species diversity, and formulate reasonable management plans (Hauer et al., 2020a, b; Ma et al., 2020a, 2020b). Long-term management strategies can be formulated to enhance ecological services and other functions (Klobucar et al., 2020) and monitor the dynamic growth of urban forests over time (Hauer et al., 2020a, b). An efficient inventory assists in planning by identifying problems and allowing managers to prescribe and assess the efficacy of potential solutions. In contrast, the absence of an up-to-date tree inventory may lead to errors in decision-making and economic losses (Yimam and Kifle, 2020). Thus, at minimum, a cursory knowledge about a tree population and a fully developed inventory system, are needed to make informed decisions.

Even though inventories are important, not all public municipal forestry programs have conducted one and managers may be unsure of what parameters are worth considering. This study aims to investigate the commonalities of tree inventories with regard to the types of data that are collected and their uses in management. To accomplish this, we first sought to identify tree inventory parameters collected globally through a review and assessment of published papers and reports within peer reviewed journals that draw on tree inventory data sources. Second, we present data from two nationwide surveys of community inventory efforts conducted in the United States and Sweden. We used descriptive and inferential statistical tests to detect differences in study variables. With the data, we also used logistic regression modelling to identify variables associated with predicting tree inventory use in the two countries which vary both in size and their histories of urban forestry and greenspace management activities. Finally, we address the following fundamental question, “What comes first, the tree inventory or the urban forestry program infrastructure?” and contrast this through the two selected countries.

2. Materials and methods

2.1. Global assessment of tree inventory systems

We conducted a systematic review of tree inventory parameters documented in published papers and reports housed on Web of Science (searching across all databases from 1850 up to 2020; Fig. 1). In addition, we did a complete search of all volumes and issues of *Arboricultural Journal*, *Arboriculture & Urban Forestry* (formerly *Journal of Arboriculture*), and *Cities and the Environment* (aka CATE) as these peer-reviewed journals are not listed on Web of Sciences, yet commonly contain tree inventory papers. The intent of this review was to ascertain tree inventory parameters that researchers collected and reported in peer-reviewed publication. Within the Web of Science portion of the review, we searched for publications that included at least one of the keywords tree, forest, park, green space, or infrastructure. Searches were further restricted to focus on study sites that included urban, city, or metropolitan regions (i.e., excluding forest inventories from locations that were not a predominately built environment). An initial list of 4128 records were examined from the Web of Science Review if they included a tree inventory with documented data collection parameters. We framed the final selection on papers that included a tree inventory as central tenet of the urban forest research. The tree inventory parameters and survey approach were summarized into eight categories (background information, condition, location, purpose, risk and maintenance, soils, structural, and survey type). Parameters and categories were developed based on lists from Miller et al. (2015) and Morgenroth and Östberg (2017).

2.2. Research approach and questionnaire design

A questionnaire was structured to ascertain the community budget

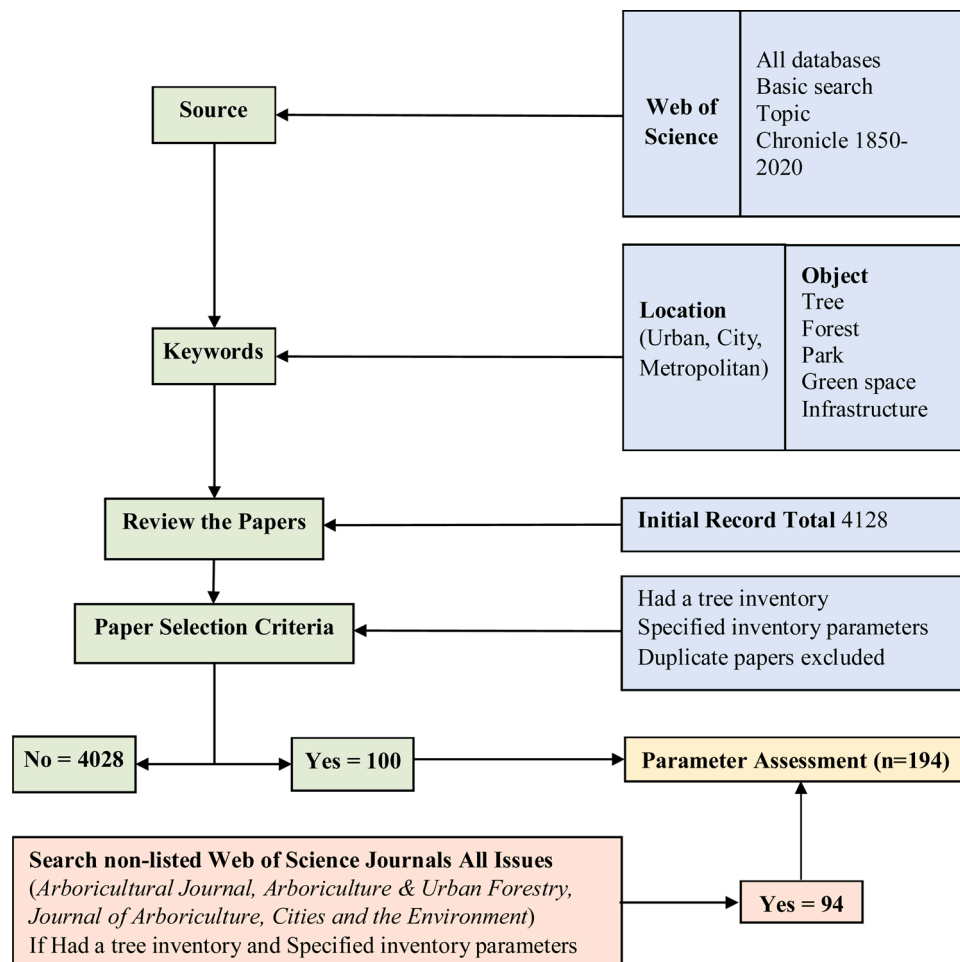


Fig. 1. The method used to obtain the global set of tree inventory systems papers published in peer-reviewed journals.

for urban forestry activities, governance structure through ordinance and engagement of the community members through a tree board, community engagement through volunteers in undertaking urban forestry activities, urban forestry strategic plans, tree management through a systematic tree management metric and adoption of tree risk management, implementation and use of tree inventory systems (Appendix A). Each questionnaire was structured specifically for each country, but the questions above were a part of and similar in each respective questionnaire. With regards to inventories specifically, we sought to determine: a) if a community had a tree inventory; b) was the inventory computerized/digital; c) what area(s) did it include (e.g., streets, parks, greenbelts, woodlots, private, other); d) what collection and analysis methods were used (e.g., windshield, sample, complete, remote sensing, canopy cover, i-Tree Streets, i-Tree Eco, other); e) who completed the tree inventory (e.g., consultant, intern, municipal staff, volunteer); f) what was collected (diameter, condition, species, insect/disease problems, planting locations, debris volume, risk assessment, removal, other; and g) how was the tree inventory used to direct urban forest management (e.g., planting, policy, removal, scheduling, species selection, pruning, canopy coverage changes). Responses were further grouped by population based on USA Census definitions with adjustment to Swedish demographics as 2500 to 9,999, 10,000–24,999, 25,000–49,999, 50,000–99,999, 100,000–499,999, 500,000 or more. We also ascertained the overall response by question and applicable question categories.

2.3. Survey design and implementation

2.3.1. Sweden survey implementation

All 290 Swedish municipalities were asked to participate in this study. A search of municipal homepages was used to develop a contact list of names and addresses of the personnel with the highest responsibility for green spaces and trees. The actual distribution of the survey and reminders was carried out following the methods of Dillman et al. (2014) and included: 1) an e-mail that was sent to all potential recipients of the survey one week ahead of the actual survey distribution, 2) a first distribution of the survey (sent electronically), 3) a reminder that was sent two weeks after initial survey to non-respondents (by e-mail and ordinary mail), 4) three weeks after the first distribution an e-mail reminder was sent to all non-respondents, 5) four weeks after the first distribution a final reminder was sent via e-mail to all non-respondents. A total 161 (55.5 %) of surveys in Sweden were returned.

2.3.2. United states survey implementation

The USA tree inventory data were collected in 2014 as part of a comprehensive, national (USA) survey of municipal forestry operations sent to 1727 communities (Hauer and Peterson, 2017). Sampling level varied by population with all communities with 50,000 or more people included and a random sample of communities below 50,000. A target 50 % of communities with populations between 25,000 and 49,999 and target 10 % of communities with populations between 2500 and 24,999 were selected. Communities were from the Community Accomplishments Reporting System (CARS) maintained by the United States

Department of Agriculture (USDA)-Forest Service. State Urban and Community Forestry coordinators from 40 states provided the primary contact for managing public trees. In cases that the primary contact information was not available, we identified a key contact person from municipal websites for the person(s) or department(s) responsible for municipal trees care. Lacking this information, an individual with senior administrative responsibilities (i.e., city clerk, city manager, mayor) was sent the survey. A multiple contact approach using methods of Dillman et al. (2014) included 1) a pre-notice letter (sent by ordinary mail), 2) the 109 question survey (with a cover letter; sent by ordinary mail), 3) a reminder email, 4) a second survey to all non-respondents (with cover letter; sent by ordinary mail), and 5) a final postcard reminder for non-respondents. A short-form version for communities who did not respond to this first survey attempt was sent as a 53-question version of the questionnaire (with cover letter; sent by ordinary mail). An email reminder was sent for this version. A total 667 (38.6 %) of surveys in the USA were returned.

2.4. Statistical analysis

Logistic regression models were developed to test variables believed to explain tree inventory adoption in the USA and Sweden (Table 1). Separate models were created for both countries. An initial global model combining both locations together was developed, but with dissimilarities in the two datasets limiting factors that could be included, preliminary modelling found separate models as more predictive. The initial USA model tested Population, Activity, BudgetPerCapita, Tree Board, Ordinance, Strategic Plan, Risk Management, Systematic Care, and Volunteer as nine independent variables to predict having an

Table 1
Variables used in study and definition.

Variable	Definition
Budget	The annual budget (USD) of the urban forestry program
Activity	Community conducted some form of urban & community forestry activities
BudgetPerCapita	The annual budget (USD) of the urban forestry program on a per capita basis
Claim	Community experienced claims for injury or property damage from public trees
Inventory	Community has a tree inventory of public trees
Inventory_Digital	A public tree inventory exists in a computer/digital format
Inventory_What	The tree attributes (e.g., diameter, species, condition) collected in inventory
Inventory_Where	Areas (e.g., street, park, greenbelt) that the tree inventory data includes
Inventory_Who	The person(s) who collected the tree inventory data
Inventory_How	How the tree inventory was used to direct urban forest management activities
Inventory_Method	The data collection and analysis system (e.g., sample, census, canopy cover) used
NetPlanted	Number of trees planted minus number of trees removed annually
NetPlantedPerCapita	Number of trees planted minus number of trees removed per person in a community
NumberPlanted	Number of trees planted annually
NumberRemoved	Number of trees removed annually
Ordinance	Community has one or more ordinances related to trees/tree management
PlantedPerCapita	Number of trees planted per person in a community
Population	Population of the community
Record	Records keep of trees planted and/or removed annually
RemovedPerCapita	Number of trees removed per person in a community
Risk Management	Community regularly conducts tree risk management
Strategic Plan	Community has a written strategic and other plan for urban forestry that includes trees
Systematic Care	The percent tree activities conducted as planned
Tree Board	Established group who reviews and recommends urban forestry management
Volunteer	Community works with partners and/or volunteers to conduct public tree activities

inventory (1 = yes). The Sweden initial model included Population, BudgetPerCapita, Strategic Plan, and Volunteer as four independent variables to predict having an inventory (1 = yes). The logistic regression was conducted using the glm() function in R (R core team, 2020) using a binomial distribution. Model simplification was conducted one term at a time, removing the least significant predictor (given p-value) and a comparing difference of fit between the original and reduced model using the anova() function as described by Crawley (2012). In the models a relative difference in fit was computed using Chi-square, ROC Curves and resulting AUC values were plotted and calculated using the ROCR package in R (Sing et al., 2005).

A chi-square test was used to determine difference between both countries with the proportion of adoption with tree inventory and other urban forestry management parameters both overall and by population group (Table 1). A t-test was used to test for difference between countries for the urban forestry budget, BudgetPerCapita, and tree planting and removal (NumberPlanted, NumberRemoved, PlantedPerCapita, RemovedPerCapita, Net Planted, NetPlantedPerCapita). A p-value < 0.05 was used to test for significant difference in all cases. Monetary SEK budget values were converted to USD using an 8.4366 SEK to 1 USD mean rate from 1/1/2015 to 12/31/2015 (Exchange Rates UK, 2021).

3. Results

3.1. Global application of urban tree inventories in research

An assessment of tree inventory types and parameters revealed differences in application in research. A total of 194 published papers were found that included an urban tree inventory as part of the research (Fig. 1). Interestingly 100 papers were found from the Web of Science review and 94 papers were found through hand searching the three non-listed journals (i.e., *Arboricultural Journal*, *Arboriculture & Urban Forestry*, and *Cities and the Environment*). Tree inventory survey types were more common as a sample (57.2 %) than population (census) survey (27.3 %), with 15.5 % of papers not listing which approach was applied (Table 2). Structural parameters were most commonly collected, with species (85.1 %), DBH (65.5 %), total height (45.4 %), and crown width (30.9 %) being the most frequently included. Soil parameters were uncommon with texture, pH, and compaction each used less than 10 % of the time.

In categorizing the research papers reviewed by intent, the advancement of planning (39.2 %) efforts was the most common underlying research objective. This was followed by ecosystem services (32.0 %), the monitoring of forest condition (20.6 %), and the assessment and mitigation of risk (19.6 %). In datasets that characterized spatial location, GPS (39.7 %), site type (35.6 %), land use (28.9 %), and address (16.0 %) were the most common variables collected. Tree condition, maintenance, or background information categories were less common with inclusion of parameters below 30 % (Table 2). Research derived tree inventory data occurred on all seven continents with the exception of Antarctica Fig. 2.

3.2. Urban tree inventory application in Sweden and the United States

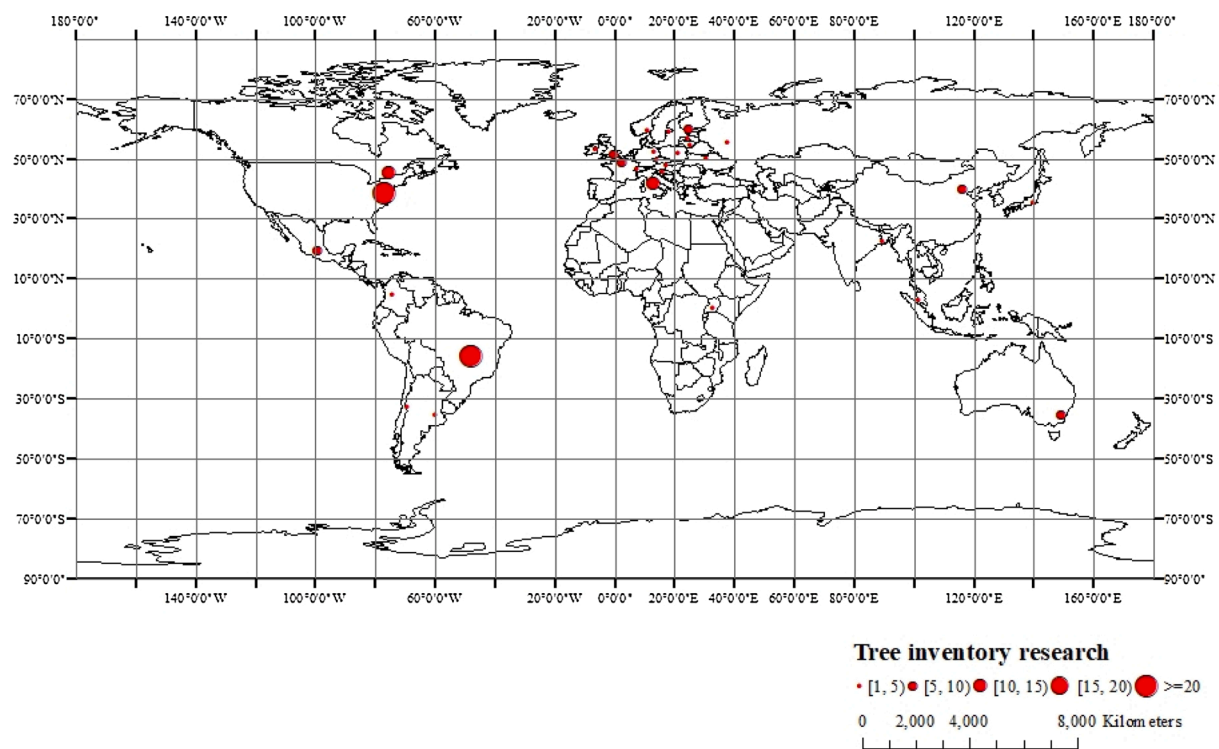
The commonality of tree inventory usage (e.g., planning, selecting planting locations, ordinance development) and components (e.g., tree board, strategic plans, systematic management) of the urban forestry programs gleaned from research questions (Appendix A) show response differences between communities in Sweden and the USA. In both countries, as population size increased communities were more likely to have a management plan ($p < 0.001$, Fig. 3), an inventory ($p < 0.001$, Fig. 4), a computer-based digital inventory ($p < 0.001$, Fig. 5), and records pertaining to tree planting and/or removal ($p < 0.001$, Fig. 6).

Overall, 50 % of the responding cities in the USA had a management plan. This was significantly higher than the 26 % rate observed in Sweden ($p < 0.001$, Fig. 3). With regard to inventories, most Swedish

Table 2

Commonality of tree inventory variables by category from a review of published reports in peer-reviewed research depicting tree inventory systems. (n = 194).

Category	Variable	Percent	Category	Variable	Percent
Background Information	Tree Record	29.9	Risk and Maintenance	Maintenance Record	10.3
	Date of Measurement	28.9		Maintenance Need	6.7
	Planting Date	8.8		Wood Condition	6.2
	Identification Code	3.1		Cost of Maintenance	3.6
	Crew Training Level	2.6		Type of Protection	0.5
Location	GPS	39.7	Structural	Species	85.1
	Site Type	35.6		Stem Diameter (DBH)	65.5
	Land Use	28.9		Total Height	45.4
	Address	16.0		Crown Width	30.9
	Actual Land Use	7.7		Height to Crown Base	7.2
	Distance/Direction	6.7		Height to Live Top	3.1
	Tree Tag	1.5		Height DBH Measured	2.1
Condition	Health Condition	28.4	Purpose	Planning	39.2
	Insect/Disease	23.7		Ecosystem service	32.0
	Overall Condition	23.7		Monitoring	20.6
	% Crown Dieback	9.3		Risk mitigation	19.6
	Mortality Status	7.7		Tree Value	16.0
	% Missing Crown	5.2		Budgeting	1.0
	Crown Light Exposure	3.1	Soils	Texture	9.3
	Crown Transparency	1.0		Compaction	4.6
Survey Type	Sample	57.2		pH	2.1
	Population	27.3		Organic Mater	1.0

**Fig. 2.** The distribution of global tree inventory research from scientific studies of urban forests. (n = 194).

communities were at or above the percentage observed in the USA, with the exception of communities with populations between 10,000–24,999 (where a greater number of communities in the USA had inventories). This largely explains the overall differences in inventory rates observed between Sweden (57 %) and the USA (67 %) communities ($p = 0.021$, Fig. 4). Digitalization of urban tree inventories was common in both countries, though significantly fewer ($p < 0.001$, Fig. 5) inventories in Sweden (63 %) were digitalized compared to those from the USA (83 %). Fewer (approximately 1 in 4 communities) in the 2,500–9,999-population group in Sweden had a digital inventory compared to the majority

(6 in 10 communities) in the USA. All responding communities over 500,000 had a digital inventory in both countries. Swedish communities were less likely to have tree planting and/or removal records ($p < 0.001$, Fig. 6). In the USA, 80 % cities had this record type and 54 % of Swedish cities had such. As cities got larger, records were common with two-thirds of communities at or above 25,000 people having records for tree planting and/or removal.

The majority of inventories were conducted by municipal staff (Fig. 7) and similar for the USA (62 %) and Sweden (58 %). Contracted consultants were also used to conduct at least a portion of an inventory

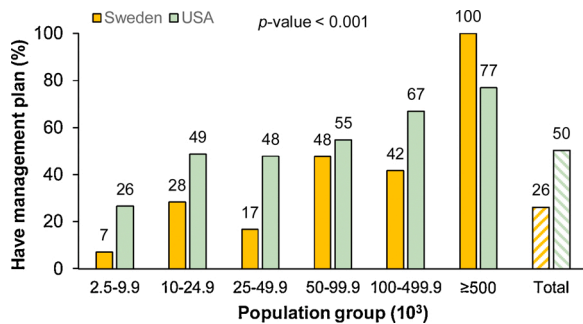


Fig. 3. Percent of communities having a management plan by population size in Sweden ($n = 145$) and the USA ($n = 594$).

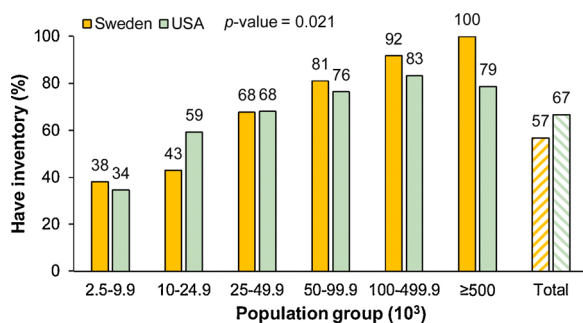


Fig. 4. Percent of communities having an inventory by population size in Sweden ($n = 150$) and the USA ($n = 656$).

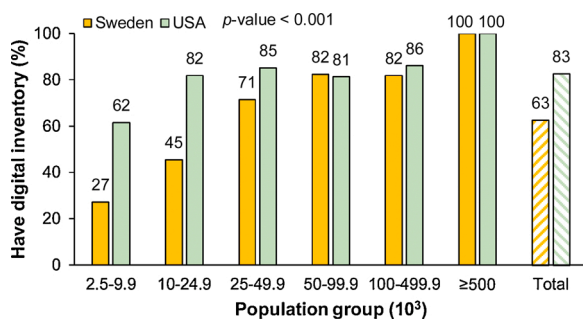


Fig. 5. Percent of communities having a digital inventory by population size in Sweden ($n = 83$) and the USA ($n = 355$).

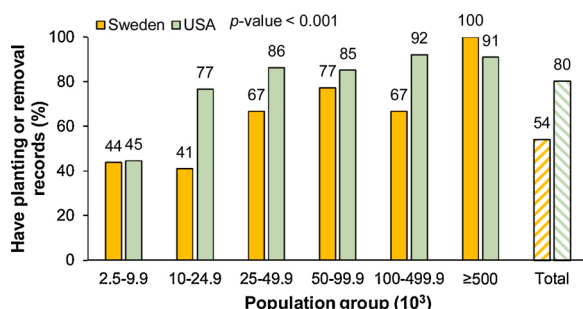


Fig. 6. Percent of communities having a record of planting or removal by population size in Sweden ($n = 161$) and the USA ($n = 493$).

in Sweden (59 %) that was greater ($p = 0.033$) than in the USA (46 %). Intern involvement was similar in the USA (24 %) and Sweden (18 %). The involvement of citizen volunteers was 14 % in the USA and greater ($p < 0.001$) than in Sweden as no responding community mentioned

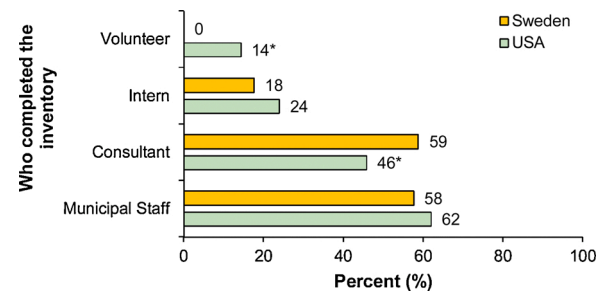


Fig. 7. People who completed the inventory by responding communities in Sweden and the USA. (*difference at p -value < 0.05).

citizen volunteers participated in any tree inventory phase in Sweden (Fig. 7) as of the study date. Volunteers were involved with other public tree activities in Sweden (13 %), but this was much lower ($p < 0.001$) than the 65 % of responding communities in the USA engaging citizens through urban tree activities.

Tree inventories occur across both public (e.g., streets, parks, greenbelts, and woodlots) and private lands (Fig. 8). Street tree inventories (94 %) were the most common and similar between countries ($p = 0.784$). Park tree inventories were regularly conducted and were more common ($p = 0.047$) in Sweden (79 %) than the USA (68 %). Public greenbelt inventories were more common in the USA than Sweden ($p = 0.005$). Approximately one-third of USA locations had had greenbelt inventories, especially in cities with increased population ($p = 0.015$). In Sweden, woodlot inventories were more common ($p < 0.001$) as population increased with approximately 1 in 4 communities having this inventory that was greater than the USA (12 %). Tree inventories that included private trees were uncommon and similar ($p = 0.121$) in both locations (Fig. 8).

Several tree inventory parameters were collected in both countries (Fig. 9). Many of these (e.g., DBH, condition, species, planting location, and debris) became more common as population increased ($p < 0.01$). But no difference was found with insect/disease, tree risk, or removal by population. Tree inventories including species information were very common and at a higher rate ($p = 0.002$) in the USA (99 %) than Sweden (93 %). In USA inventories, tree condition (88 %) and DBH (89 %) were more common ($p < 0.01$) than tree condition (77 %) and DBH (59 %) in Sweden. Collecting information for potential tree planting locations (e.g., identified vacancies) were more common ($p < 0.001$) in the USA (77 %) than Sweden (34 %). Tree removal data were similar ($p = 0.489$) in the USA (70 %) and in Sweden (66 %). Tree risk as part of an inventory was more common ($p < 0.01$) in Sweden (67 %) than the USA (51 %). Having a past tree risk claim was similar ($p = 0.579$) between Sweden (55 %) and the USA (52 %) and claims were more likely ($p < 0.001$) in larger than smaller communities. Tree debris information was similar ($p = 0.177$) and uncommon in the USA (6 %) and Sweden (2 %).

The purpose and use of a tree inventory varied between USA and Sweden (Fig. 10). Positive relationships between population and inventory uses were detected for places for planting ($p = 0.021$) and tree

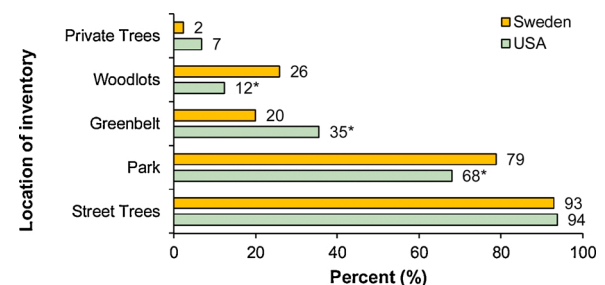


Fig. 8. Location of the inventory by responding communities in Sweden and the USA. (*difference at p -value < 0.05).

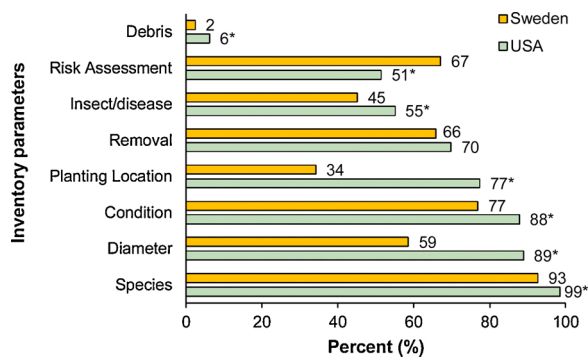


Fig. 9. Tree inventory data collected by responding communities in Sweden and the USA. (*difference at p -value < 0.01).

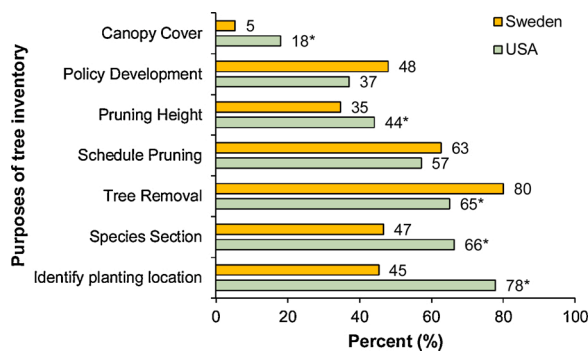


Fig. 10. Purposes of the inventory by responding communities in Sweden and the USA. (*difference at p -value < 0.01).

canopy ($p < 0.001$). As a planning tool, selection of tree planting locations (78 %) and selection of trees species to plant (66 %) was more common in the USA than in Sweden ($p < 0.001$) which were slightly below 50 % for both. The use of inventories to plan for tree removal was higher ($p = 0.011$) in Sweden (80 %) than the USA (65 %). Tree pruning (58 %) and policy setting (39 %) was similar between the two countries. The purpose of functional aspects such as ecological services through canopy cover assessment was uncommon and lower ($p = 0.006$) in Sweden (5 %) than the USA (18 %).

3.3. Parameters that predict having a tree inventory in the USA and Sweden

3.3.1. The USA regression model

The initial model found community population ($p = 0.565$), presence of urban forestry activity ($p = 0.229$), presence of a tree ordinance ($p = 0.161$), tree risk management ($p = 0.791$), and the use of volunteers ($p = 0.817$) did not explain the presence of a tree inventory and were excluded from future modelling. The final logistic regression model retained BudgetPerCapita ($p = 0.036$), having a tree board ($p = 0.013$), having a strategic plan ($p < 0.001$), and systematic tree care ($p = 0.004$) as significant variables that explained having a tree inventory (Table 3). Odds ratios ranged from 1.046 to 2.680. Communities with tree boards

Table 3

Final logistic regression with USA model variables. ($n = 382$).

Variable	Coefficient	Standard error	p value	Odds ratio	95 % CI lower	95 % CI upper
Intercept	-0.634	0.305	0.038	0.530	0.289	0.959
BudgetPerCapita (\$)	0.045	0.021	0.036	1.046	1.007	1.096
Tree Board (Y/N)	0.704	0.283	0.013	2.022	1.155	3.516
Strategic Plan (Y/N)	0.986	0.268	<0.001	2.680	1.590	4.564
Systematic Care (%)	0.012	0.004	0.004	1.012	1.004	1.020

were twice as likely to have a tree inventory as those without one. An inventory was 2.7 times more likely if a community had a strategic plan. Compared to a community being 100 % reactive to tree care (e.g., 0 % systematic care), an inventory was 3.3 times more likely with 100 % systematic care and 2.6 times more likely if a community undertakes 80 % systematic management. The final model produced an 0.723 AUC.

3.3.2. The Sweden regression model

The initial full model found BudgetPerCapita ($p = 0.335$) and volunteer (0.674) did not explain communities in Sweden to have an inventory (Table 4). The final model found population ($p = 0.001$) and written strategic plan ($p < 0.001$) as significant in predicting a community having a tree inventory. With a written strategic plan, a city was almost 12 times more likely to have an inventory than a community without a plan. Larger communities were more likely have an inventory than smaller communities. The final model produced an 0.802 AUC.

3.4. Program budgets, trees planted, and trees removed

Urban forestry programs differed in budgets, tree planting, and tree removal (Table 5). Municipal forestry budgets were significantly higher ($p < 0.001$) in Sweden (\$39.92 USD per capita) than in the USA (\$8.76 USD per capita). Interestingly, even though forestry spending per capita was greater in Sweden, fewer Sweden respondents (36 %) believed their forestry budget was adequate ($p < 0.001$) than the USA respondents (53 %) when trying to meet planned goals. Trees planted and trees removed were higher in the USA in both total number and total per capita (Table 5). However, no difference was found in both net tree planting ($p = 0.254$) and net planted per capita ($p = 0.157$) metrics.

4. Discussion

4.1. Global application of tree inventories

A tree inventory consists of various parameters that describe some aspect of the urban forest (Morgenroth and Östberg, 2017). Tree species was the most common parameter assessed, followed by DBH. The common collection of tree species was not surprising as this parameter along with diameter were important base variables for ecosystem assessments (Steenberg et al., 2017). Östberg et al. (2013) found that of 148 tree inventory parameters assessed through a Delphi study, tree species occupied the top place. We also found in this study that tree species was the most common parameter in Sweden (93 %) and the USA (99 %). In this study, diameter was also a common collected parameter

Table 4

Final logistic regression Sweden model variables. ($n = 141$).

Variable	Coefficient	Standard error	p value	Odds ratio	95 % CI lower	95 % CI upper
Intercept	-0.977	0.306	0.001	0.376	0.201	0.672
Population (#)	2.65E-05	8.30E-06	0.001	1.000	1.000	1.000
Strategic plan (Y/N)	2.475	0.649	<0.001	11.881	3.801	52.672

Table 5
Municipal forestry program budgets, tree planted, and trees removed.

Variable (significance)	Country	n size	Mean	Standard Deviation	Standard Error
BudPerCapitaUSD ($p < 0.001$)	Sweden	143	39.92	44.71	3.74
	USA	477	8.76	9.93	0.45
NumberPlanted ($p = 0.103$)	Sweden	79	89.73	168.2	18.92
	USA	349	629.5	3717.8	199.01
NumberRemoved ($p = 0.008$)	Sweden	49	78.69	164.3	23.48
	USA	344	467.5	1292.4	69.68
PlantedPerCapita ($p = 0.009$)	Sweden	79	1.60	1.71	0.19
	USA	349	7.12	20.53	1.10
RemovedPerCapita ($p = 0.016$)	Sweden	49	2.00	4.31	0.62
	USA	344	5.99	12.27	0.66
NetPlanted ($p = 0.254$)	Sweden	48	18.23	121.86	17.59
	USA	334	155.2	2939.8	160.9
NetPlantedPerCapita ($p = 0.157$)	Sweden	48	-0.37	4.31	0.62
	USA	334	1.07	22.45	1.23

in the USA (89 %) but collected less so in Sweden (59 %). The difference might be due to the focus of US municipalities to use i-Tree Eco to make ecosystem analysis of urban trees (Kielbaso, 2008; Morgenroth and Östberg, 2017; Rogers et al., 2017), where DBH was a compulsory parameter. Whereas Northern Europe has focused more on management issues, (e.g. tree health and management) related to monitoring the dynamics of the tree stands (Keller and Konijnendijk, 2012; Morgenroth and Östberg, 2017), thereby not having to include DBH in their inventories. It was also possible that differences in ecologic zones explain less tree diversity, thus lower use of this parameter in more northerly latitudes. Although some evaluators believe that stem size should be measured at 1 m or other heights (Östberg et al., 2013), the formulation in the Swedish survey “Do you collect stem diameter / stem circumference?” should most likely have avoided such biases.

In our review of global research, a tree inventory was most commonly used for planning, followed by ecological services to facilitate urban forest management. As a tool to quantify outcomes of planning and the value and ecosystem service of urban trees, a tree inventory reflects the current situation and provides a baseline to monitor changes of vegetation in a city (Hauer et al., 2020a; Hauer et al., 2020b; Podrážský et al., 2014). A tree inventory or urban forest assessment can be used to test the effect of urban forestry policy and governance decisions. An assessment of tree canopy is one way and Hilbert et al. (2019) showed a heritage tree preservation ordinance was associated with 6.7 % greater urban tree canopy in communities.

Global tree inventory data also showed some parameters (e.g., soils, maintenance need, and background information) were seldom the focus of studies. This might be that the research focus was more aligned to vegetation itself and a focus on ecological and functional research (Escobedo et al., 2015; Pinheiro et al., 2018), or maybe researchers had not yet found necessity for such soils data. In addition, the traditional investigation of vegetation information was very time-consuming and laborious in itself (Ullah et al., 2017). Likewise, collecting parameters such as soil information can also be laborious but important (Scharenbroch et al., 2005). However, soil information provides a vital link to understand and predict tree growth, but challenging to predict growth in all locations (Scharenbroch et al., 2017). As with any study or local tree inventory, not all information can be collected and the judicious selection of applicable parameters important for decision making (Miller et al., 2015). Further, there is a fixed cost of travel to a tree inventory plot or individual tree and consideration with collecting additional parameters is ideally evaluated to make certain important information is not missed and to decide if the added variable cost might warrant additional parameter collection.

4.2. Urban tree inventory application in Sweden and the United States

We found in general as population increased a greater proportion of

communities had a management plan, tree inventory, digital tree inventory, and planting and removal records. This trend was not unexpected and has been found in other urban forestry studies (Giedraitis and Kielbaso, 1982; Hauer et al., 2018; Kielbaso et al., 1988; Koeser et al., 2016; Ottman and Kielbaso, 1976). The proportion of communities with tree inventory activities were higher for several parameters in the USA than in Sweden. This might be a function of the development of urban tree management sooner in the USA than in Sweden and other European countries (Hauer et al., 2017). For example, in response to Dutch elm disease (*Ophiostoma ulmi*) in the 1960's and 1970's that severely impacted urban tree populations in the USA, the USDA-Forest service sponsored urban tree management research and demonstration projects (Hauer et al., 2008, 2020a; Sinclair and Campana, 1978). Tree inventories were one such outcome of these urban forestry capacity building assistance mechanisms in the USA. The development of urban forestry programs in Europe was assisted with baseline green space and urban forestry assessments, capacity building studies, and formal research prioritization proposals in the 1990's (Konijnendijk, 2003; Konijnendijk et al., 2006). However, only during the past few years has i-Tree Eco been commonly used in European conditions. In Sweden, urban tree inventory implementation on a single tree level was more recent with Sjöman et al. (2012) documenting such through a tree inventory comparison of 10 cities. However, a tradition of urban woodland management in Europe and related sample inventories has a longer history with focusing on stand-based approaches on the urban fringe (Rydberg and Falck, 2000). Urban tree inventory use and research was now more advanced in Sweden, but possibly the lag period between the two countries was one explanation for differences (Östberg et al., 2018).

An urban forestry management plan provides a community with a sense of what they have (e.g., inventory), what the community wants (e.g., goals), and lays out a framework (e.g., course of action) to achieve their goals (Miller et al., 2015). Formal plans in a community to address urban trees were less common in smaller communities, more common as population increased, yet only half of USA and a quarter of Sweden communities had such. One potential way to increase planned activity through management plans is through technical and financial assistance (Hauer et al., 2008). Federal and state urban and community forestry capacity building grants were successful with building local urban forestry capacity in communities in the USA (Hauer et al., 2011). These capacity building activities in the USA may partially explain higher management plan frequency in the USA than Sweden. In Sweden, funding has not been given for urban tree inventories, but instead for large scale inventories of trees with high biological values (Naturvårdsverket, 2004). The national database now contains 587,923 trees, distributed around Sweden (Artportalen, 2020). Swedish municipalities have, in comparison, not been given any state guidance or funding for urban forestry programs but a more recent goal from the government is that most Swedish municipalizes, by 2025, should have integrated ecosystem service planning in all development, construction and management in urban areas (Boverket, 2020). This suggests a clear focus on future tree management in Swedish municipalities.

Tree inventory data collection involves a variety of people groups (e.g., professional staff, consultants, citizens, interns). The implementation of a tree inventory could use one or several groups. In both countries, consultants and staff were the main people to complete the tree inventory. An intern hired as seasonal staff to conduct tree inventory activities was less common, occurring in fewer than a quarter of communities. Internships with students currently undertaking post high school education or recent university graduates provide a potential lower cost way to collect tree data and provide a relevant experience for students (Roman et al., 2018a). Yet, mentoring and verification of a student internship conducted municipal tree inventory was important and we found no papers to validate the efficacy of this inventory group. Engagement of citizens through unpaid voluntary positions to collect tree inventories provides another way to undertake data collection (Roman et al., 2018a). Due to limited budgets or professional staff time,

volunteers were often considered as a way to conduct a tree inventory survey (Dickinson et al., 2012). However, compared with professionals, the quality of data that volunteers collect is often lower (Bancks et al., 2018; Roman et al., 2018a, 2017), and targeted training should be conducted before the survey to ensure the quality and reliability of the data obtained (Bloniarz and Ryan, 1996). The overall involvement of volunteers in this study was lower than that in other groups with 14 % of communities in the USA and there was no volunteer tree inventory involvement in Sweden. Volunteer involvement in all urban forestry activities in USA was the equivalent of 714 full-time staff and their involvement occurred in communities ranked higher along a sustainability index (Hauer et al., 2018). The engagement of volunteers in Sweden communities for other tree activities (e.g., tree pruning, governmental tree groups) did occur, but at a five times lower rate than in the USA. This was in line with the results of Keller and Konijnendijk (2012) who concluded that “No volunteers were used in Scandinavia when carrying out the inventory, and no further community-engaging projects were started as a result of the inventory”. According to Keller and Konijnendijk (2012), the use of volunteers raises concerns over the validity of the inventory itself. In Sweden, there might also be a cultural reason for the lack of volunteers used in urban tree inventories. Swedes in general do not volunteer for issues that they regard as a government/municipal responsibility, whereas they spend rather much time volunteering for sport organizations (Grassman and Svedberg, 1996). This discrepancy might be due to comparably high taxes, which makes Swedish residents reluctant to help the municipality that, in the citizens views, has already received funding to manage and inventory the trees. Further, Swedish tree managers have regarded themselves as experts and unwilling to include the citizens in management and inventory projects (Grassman and Svedberg, 1996; Keller and Konijnendijk, 2012).

An urban tree inventory can occur on both public and private lands. The proportion of public tree inventories was far higher than that of private forest land, perhaps since public trees were a municipalities' responsibility for maintenance (Escobedo et al., 2015; Hauer and Peterson, 2017). As increasing the social and environmental quality of life for citizens is often a priority of decision makers, it makes sense that street and park trees were more common given close proximity to people and property and associated ecosystem services and aesthetic value (Dobbs et al., 2014; Tucker Lima et al., 2013). Tree inventories were most common along streets (over 90 %) and in parks (approximately 75 %) and less common within public green belts and woodlots. The greater rate of inventories along streets and in parks might reflect the greater abundance of people in proximity to trees for visual effect and need to maintain safety for tree risk (Koeser et al., 2016; Wei et al., 2020). Inventories and assessment of expressions on visitors' faces based on forest attributes were essential to detect improved quality of life and stress relief (Wei et al., 2019). Few communities had an inventory assessment of private trees. The irony was the vast majority of trees occur across private lands, yet this urban forest portion was least understood (Miller et al., 2015). This contradiction can be easily understood that private lands were owned by many holders in a region and integration of these lands into public plans and governance can be difficult.

Tree inventories can employ both on the ground and remote methods for assessment. The first choice for many studies was a field survey. The use of street imagery (e.g., Google Street View) and other new technologies were being tested and implemented to improve the measurement speed and efficiency (Alvarez et al., 2015; Berland et al., 2019; Berland and Roman, 2020; Chacalo et al., 1994; Evarte-Bundere et al., 2014). This might be related to the characteristics of street trees which were convenient for investigation and a common public part of the urban forest (Sreetheran et al., 2011), which not only created a community's visual landscape, but also linked other green spaces (Evarte-Bundere et al., 2014). The fast pace of urbanization globally was enlarging the area of urbanized lands that contributes to a need for assessment approaches such as satellite imagery.

The characteristics of park vegetation in a city provide a core contribution for various environmental functions of the city such as shading people during recreational activities (Evarte-Bundere et al., 2014; Jim and Liu, 2001). Green belt inventories in the USA were significantly higher than that of Sweden, while the proportion of wood lot inventories in Sweden was much higher than that in the USA. Woodland cover was generally high in Swedish cities, acting as a core component in the green infrastructure as many cities have grown into the surrounding forest rich landscape (Nielsen et al., 2017; Rydberg and Falck, 2000) and as such planted green belts might be less prominent as green space components compared to urban woodlots. This phenomenon might also be related to the historical development of the two countries, with European communities being older and public woodlots resulting from conversion of Feudal estates into public places (Hauer et al., 2017). Actually, given the high level of urban woodlands in Sweden it might be surprising that so few municipalities included woodlots in the inventory. This can probably be explained by woodlands being part of the park organization, whereas other woodlands were part of forestry operations and managed according to more or less common forestry practice including traditional forest inventories and management (Rydberg and Falck, 2000; Skogsstyrelsen, 1991; Swedish Society for Nature Conservation, 2010).

There were many parameters in the tree inventory, but their selection should be closely linked with the purpose or use. Species was the most frequent indicator in this study. This was consistent with the most commonly used indicators in the global tree inventory research studies (Table 3). It was helpful to understand the vegetation composition, species richness and diversity, and a tree species native origin to plan future tree selection for planting (Morro and Castro, 2015; Omar et al., 2018; Rudolph et al., 2017). In addition, the collection of tree diameter and condition was also very common. In many scientific investigations, diameter was among the parameters most commonly used for its simple and easy to measure characteristics that relate to many other urban forest attributes such as ecosystem services (Östberg et al., 2013). Tree condition provides a sense of the overall forest health and can be used to test for effect of planned approaches to foster urban forest health (Zarnoch et al., 2004). These three structural indicators provide a base-line structural characteristic of the urban forest.

Several management-based parameters that were commonly collected include tree planting location, trees requiring removal, and risk assessment. Tree risk assessments as a formal inclusion into tree inventories provide data to monitor the risk status of urban trees to reduce the odds of personal injury and reduce property loss (e.g. buildings, vehicles) caused by tree failure (Koeser et al., 2016; Sreetheran et al., 2011). The importance of tree risk management was revealing in that slightly over half of communities in both countries have received a past claim for damages due to a tree. Sweden communities used the tree inventory to plan tree removal more so than the USA. Tree removal was one method to abate tree risk, but it was not known if tree removal as a tree risk mitigation method was greater in Sweden, or just more commonly a stated use of the tree inventory. According to Östberg et al. (2018), risk was sometimes used as an excuse to conduct municipal tree inventories since politicians were more willing to pay for inventories that have a risk focus than e.g. an inventory of tree canopy cover. This reflects the claim by Matthews et al. (2015) that green infrastructure questions were often seen as either capital-based or risk-based. Using the tree inventory to plan for tree species selection and to identify planting locations was a greater intended purpose in the USA (Ma et al., 2020a, 2020b). One possible explanation might be that the urban forest in the USA has suffered from more pests and diseases such as Dutch elm disease, emerald ash borer (*Agrilus planipennis*) and Asian longhorned beetle (*Anoplophora glabripennis*). In comparison, Swedish municipalities have mainly been faced with Dutch elm disease, thus species selection was currently a less crucial part of mitigating future outbreaks of pests and diseases. Finally, tree canopy cover was not a common function of the tree inventory in either country, however,

nearly 1 in 5 communities in the USA did use the inventory for this purpose. It was possible that tree canopy planning was more advanced in the USA as an inventory purpose as a result of promotion by organizations (e.g., American Forests, the USDA Forest Service, and state urban and community forestry programs) through active tree canopy and ReLeaf promotional campaigns initiated in the early 1990's (Kukko et al., 2017; Riemann et al., 2016).

4.3. Parameters that predict tree inventory in the USA and Sweden

We set out to test and understand how various urban forest program attributes could explain why one community might more likely have a tree inventory than another. A question from this study sought to answer, "What comes first the inventory or the program?" We found several variables that explained a greater tree inventory likelihood. Planning was an important explanation in both countries. Thus, a community was more likely (greater odds) to have an inventory in the USA (2.7) and Sweden (11.9) if they formally developed an urban forest strategic plan, or other plan that address trees. This positive correlation between strategic plans and inventories had also been reported for green spaces in Norway (Fongar et al., 2019). Having people involved through governance was found as an important predictor as a reason for community adoption of a tree inventory in the USA model. This variable was not part of the Sweden data set, thus we do not know if governance would help explain tree inventory integration in Sweden. A community that conducted systematic (e.g., planned) tree management was several times more likely to have an inventory than places with a reactive mode of action (Table 3). Finally, having greater financial resources also accounted for greater odds of having a tree inventory (Table 3). Adequate financial ability and/or allocation priority for inventory tools, personnel, and management information systems were important for having a tree inventory. Thus, we found several programmatic factors that appear as prerequisites for having a tree inventory. The absence of one or more of these programmatic factors does not mean a community will not have an inventory, rather they were less likely to according to the models in this study. Further, it was unlikely a community develops a tree inventory as the first thing they do. Rather, initiation of basic urban forestry functions such as tree planting and tree removal, hiring staff, financial allocation, governance support, and strategic planning were found as supporting reasons why one community but not another might have a tree inventory.

Östberg et al. (2018) suggested the tree inventory and green space management budget were related and both important for good tree management. Hauer and Peterson (2017) also found the municipal tree budget was an important indicator of emerald ash borer management and the allocation for budgets for tree removal and planting. Having a tree board provides a mechanism for community governance and decision making with program prioritization, with a tree inventory one outcome. The allocation of professionals also played a positive role in promoting tree inventory implementation. Employing or retaining under contract qualified staff (e.g., ISA certified arborist) was important to conduct a tree risk survey (Hauer and Peterson, 2017). Systematic care, as the percent of tree activities conducted proactively according to a management plan, such as planting, pruning, and removal explained more advanced urban forestry programs (Hauer et al., 2018). Having a written strategic plan for the urban forest and other plans that included trees might provide an easier path to conduct a tree inventory if the plan specifies this as a goal (Gillespie, 1999).

In comparing the USA and Sweden models (Tables 3 and 4), population was related to the likelihood of having an inventory only in the latter country (Table 4). The Initial USA models found population as not significant. Rather, BudgetPerCapita better explained the presence or absence of an inventory. In contrast, in the Sweden model BudgetPerCapita did not explain if a city had an inventory. In all but the smallest communities (e.g., < 10,000 people), funding was relatively similar between communities with or without an inventory in Sweden

(data not shown). In this country, smaller municipalities tend to focus less on collecting information (Östberg et al., 2018) and larger cities had more abundant financial, material and human resources to support a tree inventory. Thus, in Sweden a lack of variability among community budgets and an emphasis on strategic plans may explain why BudgetPerCapita was not significant.

4.4. Program budgets, trees planted, and trees removed

An interesting finding from this study was an approximate 4.5 times greater per capita urban forestry budget in Sweden. This might be due to difficulties in comparing budgets since green space management at the local governmental level covers many different types of areas (Cvejić et al., 2015), but it might also have been affected by the clear goals set by the Swedish government and EPA and the implementation of 16 national environmental quality objectives (Naturvårdsverket, undated). Swedish municipalities were also actively using urban trees and green spaces to market themselves as green, such as the city of Växjö that uses the slogan "The greenest city in Europe" in much of their marketing (Växjö, 2020). Interestingly, more Sweden respondents also believed their budget resources were inadequate to meet planned goals compared to USA respondents. Overall tree planting and removal was higher in the USA. However, when normalized on a per capita basis, no difference was found between both countries. The higher tree removal and planting in the USA was likely the result of trees affected by emerald ash borer and common abundance of the host tree species (*Fraxinus* spp.) explaining the difference (Hauer and Peterson, 2017; Ma et al., 2020a, 2020b).

4.5. Study limitations

As with any study the results are reflective of the data used to test a question. In this paper we used the response of people we deemed best able to answer questions about their municipal tree programs. It is possible that a person better able to answer questions was over-looked. However, we took steps to best identify that key person to query programmatic knowledge of tree inventory systems by using both the experience of the principal investors, existing lists of key contacts, and municipal web page verification or identification of key contacts. We also used a sample to estimate the population of municipal forestry programs as pertaining to tree inventory systems and it is possible sampling bias exists. However, we took precaution with multiple requests to participate in the study and tested for non-respondent bias, finding none with the USA dataset. We used a method to identify papers published in peer-reviewed journals that address tree inventory systems interpreted to be used for urban tree population management. We further expanded the search to include three non-indexed journals that historically published urban tree inventory papers. It is possible that not every tree inventory paper was found, but the authors took precaution to reduce this risk. The inherent nature of paper selection in studies using databases such as the Web of Science by design limit paper selection by decision criteria associated with the databases that limit inclusion to a subset of information that pertains to a study question such as in this paper. In our case the Web of Science returned 100 papers and we found an additional 94 papers by manually searching every issue of three non-indexed journals as described in the methods.

5. Conclusion

This study found several indicators that explain why a community has a tree inventory. Having a strategic plan that likely explained the importance of a tree inventory and what to collect was paramount for adoption of a tree inventory in a community as found in both countries. Urban forestry program development and sustainability were likely predicated on having sufficient capital resources (e.g., an adequate tree budget), community representation through governance (e.g., a tree board in the USA), and a program advanced by practicing active

management (e.g., systematic care in the USA). Thus, what came first the tree inventory or the program? Findings from this study suggest some base infrastructure (e.g., tree budget, tree board, and systematic care) was important for explaining a higher likelihood of a tree inventory being in a community. While no one attribute assures that a tree inventory will occur, each helps explain why some communities choose to inventory their trees. A community ultimately decides the value that occurs from having knowledge about a tree population. This knowledge will likely help with planning and management of urban forests and green spaces.

We also identified the abundance of various tree inventory parameters. Using results from both Sweden and the USA along with a global summary of 100 research papers, tree species was the most frequent parameter in the tree inventory, followed by tree diameter and tree condition. Staff with professional knowledge were common collectors and an important part with the quality of the tree inventory. The inclusion of volunteers occurred in few communities and their inclusion serves to engage the public. This could also perhaps decrease the cost of data collection in communities that face budgetary challenges, assuming adequately trained volunteers results in data that meets or exceeds collections standards. Finally, we identified tree inventories in street and park locations (both places with more people per tree) were more frequent than green belts and woodlots (places with fewer people per tree). Collection of tree risk ratings in inventories that address potential risk of injury and damage was important and could reduce tree related claims which were common to about half of the communities.

CRedit authorship contribution statement

Bingqian Ma: Conceptualization, Formal analysis, Data curation, Writing - original draft, Writing - review & editing. **Richard J. Hauer:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Johan Östberg:** Conceptualization, Methodology, Investigation, Writing - review & editing, Project administration, Funding acquisition. **Andrew K. Kooser:** Conceptualization, Methodology, Formal analysis, Writing - review & editing. **Hongxu Wei:** Funding acquisition, Conceptualization, Writing - review & editing. **Chengyang Xu:** Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

The authors have no conflict of interests associated with this research paper.

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Appendix A. Questions Used to Ascertain Urban Forestry Program and Tree Inventory Systems Implemented in the Community

- 1 Did your community conduct any kind of shade tree/urban & community forestry activities? (Yes/No)
- 2 What is the population of your incorporated community area? (#)

- 3 What is the total annual budget of your municipality funded tree care activities and management from all municipal sources? (Include all tree activity expenses; include personnel, overhead, equipment, supplies, tree care and contract payments.) (#)
- 4 Does your community have a government-authorized tree board, parks board, city department, commission, or similar group that helps develop and/or administer tree management policy? (Yes/No)
- 5 Does your municipality have one or more municipal ordinances that pertain to trees? (Yes/No)
- 6 Does your community have a written strategic plan for urban forestry, tree management, open space, green infrastructure, or land use management that includes trees?
- 7 Does your community work with partners and/or volunteers (individuals or groups not paid for providing services) for tree planting, tree care, or other tree activities on public property? (Yes/No)
- 8 Does your community have a tree inventory? (An inventory is any type of record of public trees in your community.) (Yes/No)
- 9 Is your inventory computerized/digital? (Yes/No)
- 10 What areas does the tree inventory include? (Yes/No)
 - a Street trees
 - b Park trees
 - c Municipal greenbelts
 - d Municipal woodlots
 - e Private trees (sample survey)
 - f Other municipal properties (please specify: _____)
- 11 Which type of inventory collection and analysis methods have been used to describe your community tree population?
 - a Windshield survey
 - b Sample survey
 - c 100 % population (total, census)
 - d Remote sensing (i.e., aerial photos, satellites)
 - e Canopy cover analysis
 - f i-Tree Streets analysis
 - g i-Tree Eco analysis
 - h Other (please specify: _____)
- 12 Who completed your tree inventory? (Yes/No)
 - a Consultant
 - b Intern
 - c Municipal staff
 - d Volunteer
 - e Other (please specify: _____)
- 13 The tree inventory is used to identify which of the following? (Yes/No)
 - a Tree diameter
 - b Tree condition
 - c Tree species
 - d Insect/disease problems
 - e Tree planting locations
 - f Potential debris volume
 - g Tree risk assessment
 - h Tree removal
 - i Other (please specify: _____)
- 14 Is your tree inventory used to direct any of the following urban forest management activities? (Yes/No)
 - a Identifying tree planting locations
 - b Policy and ordinance development
 - c Removal of trees exceeding acceptable risk rating
 - d Scheduling tree pruning
 - e Selection of tree species for planting
 - f Tree pruning for height clearance (street/sidewalk)
 - g Assessing canopy cover change between time periods
 - h Other (please specify: _____)
- 15 Does your community regularly conduct tree risk management (hazard tree identification)? (Yes/No)

- 16 Has your community experienced any claims for injury or property damage from public trees? (Yes/No)
- 17 How many trees are planted annually? (#)
- 18 How many trees are removed annually? (#)
- 19 Did you keep records on the number of trees that were planted/removed? (Yes/No)
- 20 What percent of tree care (pruning, pest control, etc.) is done on a systematic (regularly scheduled) cycle and what percent on demand as reactive (complaints, hazardous situations, crisis, post storm etc.)? (Total = 100 %)

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